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INTER PACT

UNITED STATES

INFORMATION WARFARE: WAGING AND WINNING CONFLICT IN CYBERSPACE K1

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Information Warfare:^(tm)

Waging and Winning Conflict in Cyberspace

An Outline for a National Information Policy

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An Introduction to Information Warfare

At one point, if not already, you will become the victim of Information Warfare.

If not you, then a member of your family or a close friend.

Your company will become a designated target of Information Warfare. If not yesterday or today, then definitely tomorrow. You will be hit.

And this is because the United States is at war, a war that most people have not bothered to notice. The 20th century information skirmishes which are the prelude to global Information Warfare have begun.

Information Warfare is coming, though for some, it has already arrived.

As the specter of global world-ending warfare recedes into the history books, (and hopefully stays there!) a collective sigh of complacency is replacing bomb-shelter hysteria. Despite the fact that nearly 175 million people were killed in the 20th century from the effects of war and politics, Strangelovian predictions thankfully never came to pass. However, as equally dangerous international economic competition supplants mega-tonnage intimidation, offensive pugnacity will be aimed at the informational and financial infrastructure upon which our Western economy depends.

The Cold War is over and has been replaced by Economic Warfare - a competition between what is shaping up to be three major trading blocks; North America, Europe and the Asian Pacific Rim. Richard Nixon was fond of saying in the 1970's and 1980's that World War III had already begun and that it was an economic war; perhaps one that the United States was destined to lose. In retrospect, we might have been more attentive to his prescience.

These three huge economic forces accounts for about one quarter of the population and 80% of the GNP of planet Earth. The stakes are enormous and everyone wants a piece.

The foundation of modern society is based on the availability of and access to information that will drive a thriving economy or propel a weak one into a position of power. In today's electronically interconnected world, information moves at the speed of light, is intangible, and is of immense value. Today's information is the equivalent of yesterday's factories, yet considerably more vulnerable.

Right now, the United States leads the world into a Globally Networked Society. A true Information Age where information and economic value become nearly synonymous. With over 100,000,000 computers inextricably tying us all together through the most complex array of land and satellite based communications systems, a major portion of our domestic \$6 Trillion economy depends upon their consistent and reliable operation.

Information Warfare is an electronic conflict where information is a strategic asset worthy of conquest or destruction. Where both computers and information systems become attractive first strike targets. A well financed and motivated adversary could undoubtedly bring the economy of the United States to its knees.

As I told a Congressional Committee on June 27, 1991, "Government and commercial computer systems are so poorly protected today, that they can essentially be considered defenseless. An electronic Pearl Harbor waiting to happen. As a result of inadequate security planning on the part of both the Government and the private sector, the privacy of most Americans has virtually disappeared." (1)

Computers in Crisis, "a report published in October of 1990 by the National Research Council clearly echoed my sentiments. "The modern thief can steal more with a computer than with a gun. Tomorrow's terrorist may be able to do more damage with a keyboard than with a bomb." (2) Maybe we are finally beginning to hear the warnings. In a recent study, 2/3 of Americans polled said that computer usage should be curtailed if their personal privacy was at risk. As a country, we are only now beginning to accept the fact that our personal and economic interests are indeed merging with our national security interests.

Information Warfare is an integral component of the new economic and political world order. Economic battles are fought and will continue to be fought; ultimately affecting every American citizen and company as well as the national security of the United States. As terrorism now invades our shores, we can expect attacks not only at airliners and water supplies, but upon the money supplies - a sure way to strike terror in millions of people with a single keystroke.

Since World War II the United States has "based its defensive position on our adversaries' capabilities, not their intentions." Voila! La Arms Race. However, we have not kept up with the 'Joneses'. The world is moving into Cyberspace, but our national defensive posture is still firmly landlocked.

Cyberspace is a brave new world that only luminaries such as McLuhan and Arthur Clark glimpsed in their mind's eye, but not even they could presage the uncertainties unleashed in the last two decades.

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Imagine a world where information is the commodity of exchange and cash is only used for pedestrian trade. Where information, not English, German, Japanese or Russian, is the common language. A world where the power of knowledge and information usurp the strength of military might. A world totally dependent upon its new high-tech tools to make information instantaneously available to anyone, anywhere, at any time. A world where electronic privacy is nearly non-existent.

Now imagine a conflict between adversaries where information is the prize, the spoils of war. A conflict with a winner and a loser. A conflict where computers are turned into highly effective offensive weapons. A conflict where both computers and communications systems are primary targets and must defend themselves against deadly, invisible bullets and bombs.

Imagine rival economies battling for a widening sphere of global influence over the electronic financial highways where no expense is spared to assure victory.

Then imagine that companies settle disputes and compete by regularly blitzkrieking each other's information infrastructure. A world where electronic and competitive espionage are the expected manner of conducting business. The norm.

Or imagine where personal revenge, retribution, getting even, is only a keystroke away.

What kind of world is this?

This is the world of Information Warfare. And we, as Americans and as a country, are not prepared.

Information Warfare is about how Information Age Weaponry is replacing bombs and bullets. The weapons of Information Warfare are no longer restricted to the Government or the CIA or KGB. Anyone can buy them - most of them available through mail order if you cannot find them at Radio Shack. Computer and communications weapons are available from catalogs, retail store fronts, trade shows and many can be built from hobbyist parts at home. And of course the military is developing its own arsenal of weapons to wage Information Warfare.

Information Warfare is about money. The acquisition of wealth, and the denial of wealth to competitors through economic or industrial espionage breeds Information Warriors who battle across the Global Network in a game of Cyber-Risk.

Information Warfare is about power. He who controls the information controls the money.

Information Warfare is about fear. He who controls the information can instill fear in those who want to keep their secrets a secret. Or the fear that the Bank of New York felt when it found itself \$23 Billion short of cash in only one day.

Information Warfare is about arrogance. Like the 2 teenagers who electronically vandalized a computer system to the tune of \$5 million because the company forgot to ship the boys promotional T-shirts.

Information Warfare is about politics. When the German government sponsors intelligence agency hacking against U.S. computers, the concept of 'ally' needs to be redefined. Or when Iran aims at the U.S. economy by state-sponsored counterfeiting, we should have a glimmer that conflict is not what it once was.

Information Warfare is about survival. France and Israel grew industries and their respective economies by stealing American secrets. Japan and Korea purloin American technology as it comes off the drawing boards with the help of their governments.

Information Warfare is about defiance and the disenfranchised masses in both modern and third world societies. From inner city hackers who have nothing to lose, knowing the prospect of jail represents an improvement in their lives, to gangs to organized crime, they and a host of fringe elements are seeing the cost benefits of Information Warfare.

Information Warfare is about control of the information, and we maintain less and less control as Cyberspace expands and electronic anarchy reigns.

Given global conditions of the late 1980's and 1990's, Information Warfare is inevitable and to varying degrees, the skirmishes have already begun. Today's planet offers ripe conditions for Information Warfare, many of which could not have been foreseen even a few short years ago.

Information Warfare is currently costing the United States an estimated \$200 - \$300 Billion per year, and the financial impact on our economy increases every year. Almost 5% of our GNP slithers through the Global Network and out of our control, thereby hurting deficit reduction efforts, severely impacting our export base and thus the current trade imbalance.

But \$300 Billion is mainly about people. 9,000,000 Americans who aren't working because they, too, are the victims of Information Warfare. For every \$1 Billion in commerce, about 30,000 people stay employed. For every \$1 Billion we do not trade, about 30,000 people lose their jobs. \$100 Billion alone would put 3,000,000 Americans back to work.

With \$300 Billion less in commerce, lower taxable revenues and taxable assets deprive the Government of its fair share of profits. Again, a direct economic effect on our citizenry.

As a country, more than our image is tarnished by our constant place as victim in the Information Wars. Our credit is less credit-worthy; our ability to buy and trade suffers; our political and diplomatic impact is reduced because our economic strength is no longer the pinnacle of respect it once was. We're not the only tough guy on the block anymore.

Information Warfare takes advantage of our reliance, indeed our addiction to automation and modern computerized niceties. Information Warfare attacks our very way of life.

Until pro-active defensive measures are taken, any company or organization can be severely handicapped or fatally crippled if its information infrastructure is successfully attacked. If a motivated, reasonably well financed adversary attacks strategic portions of its opponent's communications and information processing systems, such efforts can trigger an economic collapse.

Depending upon the primary target, economic dominoes can fall and severe secondary damages can result. The promise of a future Computer Chernobyl is not empty. It's only a question of who and when.

Waging Information Warfare is for anyone with an agenda and an attitude, and can be waged at three distinct levels of intensity, each with its goals, methods and targets.

Against persons or individuals.

Against organizations, companies and groups.

Against entire econo-political blocks or spheres of influence.

Class 1: Personal Information Warfare

There is no such thing as electronic privacy. The essence of who we are, our very being, is distributed across thousands of computers and data bases, over which we have little or no control. From credit reports to health records; from DMV computers to court records to video rentals; from law enforcement computers to school transcripts to debit card purchases; from insurance pro files to travel histories to our personal bank finances; every thing we do and have done is sitting somewhere in a digital repository.

The sad fact is that these very records which define us as an individual remain unprotected, subject to malicious modification, unauthorized disclosure or out and out destruction. For \$25 per name, Social Security Administration employees sold our innermost secrets. Worse yet, as of today, there is nothing you can do to protect the 'Digital You.' You have never been given the option or the opportunity to keep you and your family protected from electronic invasions of privacy.

Your life can be absolutely turned upside down if the Digital You ceased to exist. Electronic murder in Cyberspace: You're just gone. Try proving you're alive; computers don't lie. Or if the picture of the Digital You is electronically redrawn just the right way, a prince can become a pauper in microseconds.

The privacy we all consider sacrosanct has disappeared and most of us probably don't even know how or why this has happened.

Class 2 : Corporate Information Warfare

Corporate America is in much worse shape, and does not even realize that with a few well chosen keystrokes, an entire company can be decimated.

Their computers and communications systems have few defenses against attack by competitors, foreign interests or their own employees. Most organizations rely upon their information systems to function and have massive sums invested in proprietary information, yet do little to protect that investment.

Corporate management has little feel for just how weak and defenseless their corporate assets have become. While the wealth of corporations is measured in the timeliness and value of their information, no company puts Information Assets on its balance sheet. Yet, without that information, the economic stability of that company is called into question. Putting a company out of business by attacking their information systems will soon become a common method of economic competition, political retribution or social commentary. The weapons and techniques of Information Warfare are as common as spread sheets and calculators.

Corporate Security will soon face the challenges of Information Warfare and will be required to defend against it, merely to insure the survival of the company. Others will ignore the warnings and Binary Bhopals will result; potentially putting thousands of people out of work and costing our country billions in damage.

Corporate board rooms often choose to protect themselves against the statistical probability that a tornado will blow away their operations centers. The one in a million chance that a flood will rage through downtown Denver prompts companies to build underground vaults expected to survive a direct 50 megaton hit.

What companies have not prepared themselves for, however, is a well organized offensive assault against their information systems - not by Mother Nature, but by Man. In most cases, management has never even considered the idea. They have unintentionally placed their companies, their stockholders, their employees and a portion of the U.S. economy in needless peril. As you might guess, the situation is not improving.

We shall discover that it is difficult to indict Corporate America alone on all of these counts. The last 15 years of spiraling growth in information processing has been and is an ad hoc revolution driven by endless heady technical world-shaking successes and evangelical visions. Diligence in weighing the risks associated with placing our entire faith on a technical infrastructure was and is in short supply.

And as we shall see, the Federal Government must shoulder much of the blame for the current defenseless posture we enjoy. In fact, it is often not in the Government's best interest to assist us in protecting our computers and networks. Their contradictory Never-Say-Anything policies of denial have caused severe harm to efforts now under way to enhance personal privacy and commercial national economic security.

Nonetheless, inane antique policies continue unabated, and in some cases, overt attempts on the part of the Federal Government have further undermined the electronic privacy of every American citizen. Their efforts, especially those of the 12 Reagan-Bush years, so far have been successful. Even President Clinton's proposal to address personal privacy and protect American businesses was met with near universal derision, suspicion and doubt. No matter how hard they try, politicians just don't get it.

Class 3: Information Warfare

The Economic National Security of the United States is in deep jeopardy because collective Capital Hill and White House wisdom has not yet realized that information is a vital National Asset. They still think exclusively in terms of military throw weight, oil reserves, Japanese cars and illegal aliens.

Outside of a forward thinking few within the bowels of the Pentagon and related intelligence services, National Security assets are viewed as those tangible items with a concrete, quantifiable and replaceable value. Information, on the other hand, is intangible and does not have an immediately quantifiable monetary worth. Unless you lose it. Then it costs a great deal more than you ever thought.

The military has spent considerable sums of money and resources to protect what they view as strategic information; construction details of the H-Bomb, war time contingency plans, real military strength or even the alleged existence of extra-terrestrials. Yet even with the efforts they put forth to protect such information, leaks and spying proliferate.

HUD, the IRS, Congress, the Social Security Administration - not to mention the Military - and most of the huge Federal bureaucratic machinery are all dependent upon their information infrastructure to provide services to American citizens. The Pentagon spends in excess of \$20 billion per year on computers and communications to insure military preparedness, but if those systems cease to function, the lives of our soldiers are immediately jeopardized.

If there were either an accidental or intentional collapse of the outdated IRS computers, or VA computers, economic and social chaos would prevail.

The specter of Information Warfare is neither an attractive nor popular vision, but one that becomes increasingly likely as the world realigns itself for the future.

We must ask, though, why will Information Warfare be fought? Is it a foregone conclusion? A necessary component of our future? The answers are timely and unique to the Information Age, but will ring bells of recognition. Information Wars are inevitable given our place in history:

1. The incredibly rapid proliferation of high-quality, high performance electronic information systems created the Global Net work - Cyberspace - thus redefining how we conduct business. Not only did business and government buy into technology, but tens of millions of us were, within less than a decade, suddenly empowered with tools and capabilities theretofore limited to a select few. The universal availability of comparatively simple technology which can be used offensively is a prime requirement for Information Warfare and Technical Anarchy is the result.

The Global Network is an historically unprecedented highway system that defies borders and nationalism. The Global Network places the keys to the kingdom, to our wealth and our digital essence within equal reach of everyone with a computer.

2. While we as a planet recede from a bipolar militaristic stand off, we unexpectedly find ourselves with dozens of new nation- states filled with unique nation-state histories each competing for its own identity. The failure of communism did not, nor does it, automatically mean that our system of democratic capitalism wins and every newly created nation-state will adapt it as its own. There are other alternatives, and not all of them are compatible. Self-interest rides high in the early part of this decade.

The rules of the game, the economic and political influence games, aren't the same for all of us. We, as Americans, play by an old rule book, where goodness, mom and Apple Pie define our competitive ethos. No one else sticks to the outmoded Puritan ethic by which we won the Industrial Revolution.

Some groups will willingly beg, borrow or steal what they want in any way that they can. Others will resort to physical violence in the pursuit of their agendas, but we just don't work that way. America and Americans are still often viewed as spoiled brats in a self-indulgent land where instant gratification counts for everything, and that image makes us inviting targets.

3. Only 25% of the planet can be considered developed, or First World, leaving several billion inhabitants in the unenviable position of being the 'Have Nots.' The 'Haves' are the comparatively rich countries in Western Europe, Japan, some of the Pacific Rim, and of course North America. The 'Have Nots' are everyone else. With the Global Network pouring avalanches of information across the borders of the 'Have Nots' in the form of text, sound and especially pictures, very quickly the 'Have Nots' want to become 'Haves.' Through CNN and Dynasty and upscale sitcoms and global programming, the 'Have Nots' see for themselves how we, the Other Half, lives, and they want their share of the pie.

When there's nothing to fear, nothing left to lose, and the only way is up, going after the King of the Hill, America, is an obvious tactic.

4. Greed is in no short supply and no individual, business or country is exempt. Businesses and governments constantly strive for advantage over each other, often relying on less than legal techniques to gain an edge. With the Global Network in place, and the proliferation of technology for every-man, greed has found its way into the fingers and keyboards of people who might otherwise never commit a crime. Greed operates at all levels, and due to the vulnerability of most information systems, provides ample opportunity to exploit their weaknesses for stupendous profits.

Waging Information Warfare effectively can be of tremendous financial value to the winner, and be devastating to the loser.

5. The effects of Information Warfare are unique in the annals of conflict. Info-Wars can be fought by remote control, invisibly comfortable behind a keyboard ten thousand miles away. No longer is it necessary to physically intrude upon the intended victim's turf. The Global Network offers a million points of entry.

Computer Terrorism mentality allows indiscriminate damage to affect millions of people from a single strike, sowing fear, suspicion and doubt. Information Warfare provides a low-budget, high tech vehicle for mass destruction.

6. Information Warfare is A Low Risk/High Reward endeavor. The odds of getting caught are low; of being prosecuted lower still and of being convicted almost nil. On the international front, countries cannot agree what to do with nuclear weapons much less an Information Warrior sitting behind a keyboard.

7. Essentially, we don't trust computers. They are out of our control, process information far too fast for us to comprehend, and most if us don't have a clue what goes on inside of them. Yet we need computers to sustain our society.

Information Warriors leverage Binary Schizophrenia, Digital Addiction and Approximation Anxiety, our inherent fear and distrust of computers, to their advantage.

8. Lastly, and perhaps most importantly, Information Warfare will be waged Because It Can. History clearly shows that any new technology, regardless of its original intentions, soon enough finds its way into the arsenals of the warriors - in this case, the Information Warriors.

Information Warriors come in all shapes and colors. On the global front, Japanese and their cameras can represent the equiv alent of the Army Scout providing headquarters with valuable strategic information. Hackers and Phone Phreaks have been waging mild Information Warfare against Corporate America and the phone companies for years, but the recent generation of young Cybernauts is more aggressive - patently echoing the ills of society as a whole.

The Soviets, of course, were Information Warriors par excellence, and now tens of thousands of ex-Iron Curtain intelligence agents seek to ply their trade for the highest bidder; some going as far as offering their services in the classified sections of daily newspapers.

Power hungry dictators, radical fundamentalists and a score of international political sects each have reason to use Cyberspace as a means of effecting their policy. The narco-terrorists are well financed, well armed, and have a bevy of technical advisors in their stable of resources. These groups have already aimed at the Drug Enforcement Agency, costing the U.S. billions in Cash-to-Colombia and lost worker productivity.

Environmental groups have shown the willingness to be physically provocative, and Information Warfare offers them the ability to strike out at logging camps or anti-bird groups or the oil companies in a new, imaginative and less dangerous fashion.

Information brokers and data bankers sell your name, your upper-middle-class Zip code and the last time you bought underwear to anyone with a floppy disk. All without your permission. Banks and credit bureaus allow computers to make decisions that affect our lives and our livelihoods based upon information that contains as much as 30% erroneous data. All with virtual impunity.

Anyone can be an Information Warrior. Publications such as *2600 - The Hacker Quarterly*, *Phrack* and other magazines provide the basic training for inductees. Cyberspace itself offer safe havens for Information Warriors to build their armies, develop their weapons and deploy them. An unhappy worker can suddenly turn against his employer with little chance of prosecution. A government employee may moonlight as an Information Warrior or a teenager may live in Cyberspace 20 hours a day, alighting only for Coke and Pizza.

Potentially, a hundred million Information Warriors are out there, waiting, honing their skills, some less friendly than others. Information Warfare is about capabilities, the power of the individual and the power of an organized group. The capabilities of kids, the capabilities of technical mercenaries, and the capabilities of nation-states are a threat to us all. Their intentions are secondary. The point is, if any one group wanted to wreak havoc, today, they have the means, the tools, the weapons to do exactly as they please.

As we learn more about Information Warfare, we will have to answer questions that most of us have not even bothered to ask yet.

What will Information Warfare look like? How will we recognize it? How will it be waged? Who are the Information Warriors? Where are they? What do they want? Who will be pushing the buttons? What are the weapons used in waging Information Warfare? What can they do? Where can you buy them? How do they work and how many people know how to use them? How much damage can they really do? What is the Pentagon and other military organizations doing to prepare for upcoming Information Wars?

And then, why are we in this mess? Who is to blame?

Why have we let the situation, our vulnerability, reach the stage it has? What were Congress and the White House thinking or not thinking about? Why are we so defenseless? Why did they keep it a secret? Is there someone to blame? Is an electronic Pearl Harbor just a round the corner? What do we have to do to protect our National Economic Security?

This book presents disturbing answers to some very simple questions about our personal, corporate and national vulnerability as the victims of Information Warfare.

But it also provides some hope, a way out of the technocratic quagmire in which we find ourselves immersed. The first step to reparations is the admission of the problem. With an awareness of the myriad conundra surrounding personal electronic privacy, and national economic vulnerability as a starting point, we will see that the way out is not a distant goal, never to be realized. Rather, technical solutions abound. Personal electronic privacy can be achieved and national economic security is possible - if we think it's important enough to address.

Perhaps most disturbing is why American information systems - and thus America itself - is so vulnerable to assault. It's the political machinery and outdated intelligence mechanisms, replete with paranoia of a post World War II mind set that perpetuates and deepens the chasm in which we find ourselves. We find that overzealous government agencies can at least be partially blamed. Information Warfare has been fought by spies, spooks and goblins throughout the Cold War. The

exchange of Information, the barter of data and selective release of real or ersatz information has been the name of the game and the fodder for a generation of spy novels. Information used to be relegated to a select few who operated in the name of National Security. Today, anyone with a modem and a PC can effectively compete for and have access to the same data.

The intelligence agencies like the FBI, NSA and CIA are experts at waging some types of Information Warfare and the Pentagon has its share of experts for Cold War type Information Warfare. But for 'security' reasons, the hows and whys of Information Warfare were never told to the rest of America. I do not mean this to be an absolute indictment of Washington, but certainly their compartmentalized need-to-know policies kept a lid on the true scope of the problem. Indeed, Corporate America has been and is still playing ostrich-in-the-sand and refuses to accept the fact that sooner or later, they too will be a victim of Information Warfare.

But the social framework of this country, complacency, arrogance and self-gratification contribute significantly to why we sit here, a plump duck, on the Shooting Gallery of the Global Net work. We've done it to ourselves.

Board rooms and stockholders look for net quarter results, sacrificing long term investment and future-thinking. Americans are the worst savers on the planet because all they can think about is where the net meal, mortgage payment or video game is coming from. Addiction to computers breeds anti-social sentiments, an inability to get along with real people. The technically addicted suffer anxiety attacks when the ATM machine is broken, the data base is closed or the phones are down. We've already forgotten what it was like only a decade ago.

Cyberspace, anarchy and all, is a new place to live, and one way or another, we're all moving in. We might as well figure out how to get along, since both our individual successes and national strength depend upon it. We had better learn what the rules are and be prepared to take to defend ourselves.

Yes, this book is about Information Warfare, how it's waged, who is and who will be waging it. It's also about the whys; why Information Warfare is a necessary part of our technical evolution, especially in these troubled international times.

And, it is about a solution - a proposed solution where we take control of our electronic destiny and consciously decide to stop 'winging it' through Cyberspace. Where we as Americans democratically decide whether or not we want personal electronic privacy, and how we want it administered. Where stockholders and investors can hold Corporate Management and banks accountable for dereliction of duty in Cyberspace: not protecting the information. Where this country, as a governed nation state, decides to live up to its promise: defense through strength.

That solution is a "National Information Policy." There are no rules, no common set of accepted behavior that delineates right from wrong, good from bad, or legal from illegal regarding our activities in cyberspace. We really don't even know what information is, yet our economy is based upon it. We futilely attempt to jury-rig existing old and sometimes archaic guidelines, rules and laws which simply will not work in cyberspace.

I find that most of us don't even know what questions need to be asked to create a "National Information Policy" - and in Washington that is especially true. Cyberspace is not a place over which political control can be exercised. It is, in its purest, absolute Technical Anarchy, and anarchists implicitly reject government control preferring self-restraint.

Unfortunately self-restraint and moral responsibility are not trademarks of the last 20 years, and as technology and information further intertwine with our existence, rules are required. But even before the rules, must come the ethos and morals, and before the ethos must come the concept.

The "National Information Policy" I offer is not a legislative proposal for Congress to debate, but instead a series of substantive issues and questions that must be asked, considered and answered satisfactorily before we can live in the Information Age with any sense of security, stability or trust. The "National Information Policy" I offer provides an outline by which to create a foundation for the future: A Constitution for Cyberspace.

Information Warfare is real, and it's about time we knew what we're up against.

1. Submitted testimony, Subcommittee on Technology and Competitiveness, Committee on Science, Space and Technology, U.S. House of Representatives. June 27, 1991.

2. Computers At Risk: Safe Computing in the Information Age. (National Academy Press, 1991.) p. 7